

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(trend)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(Kendall)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone

```
''' r
#1

EPA_2010 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
stringsAsFactors = TRUE)
EPA_2011 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
stringsAsFactors = TRUE)
EPA_2012 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
stringsAsFactors = TRUE)
EPA_2013 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
```

```

stringsAsFactors = TRUE)
EPA_2014 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
stringsAsFactors = TRUE)
EPA_2015<- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
stringsAsFactors = TRUE)
EPA_2016 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
stringsAsFactors = TRUE)
EPA_2017 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
stringsAsFactors = TRUE)
EPA_2018 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
stringsAsFactors = TRUE)
EPA_2019 <- read.csv(here
("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
stringsAsFactors = TRUE)

GaringerOzone_combined <- bind_rows(EPA_2010,
EPA_2011, EPA_2012, EPA_2013, EPA_2014,
EPA_2015, EPA_2016, EPA_2017, EPA_2018, EPA_2019)

```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3

GaringerOzone_combined$Date <- as.Date(GaringerOzone_combined$Date, format = "%m/%d/%Y")
class(GaringerOzone_combined$Date)

```

```
## [1] "Date"
```

```

# 4

GaringerOzone <- GaringerOzone_combined %>%
  select(Date,
    Daily.Max.8.hour.Ozone.Concentration,
    DAILY_AQI_VALUE)
summary(GaringerOzone)

```

```
##      Date      Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## Min.   :2010-01-01   Min.   :0.00200                Min.    : 2.00
## 1st Qu.:2012-07-03   1st Qu.:0.03200                1st Qu. : 30.00
## Median :2015-01-04   Median :0.04100                Median  : 38.00
## Mean   :2015-01-01   Mean   :0.04163                Mean    : 41.57
## 3rd Qu.:2017-07-02   3rd Qu.:0.05100                3rd Qu. : 47.00
## Max.   :2019-12-31   Max.   :0.09300                Max.    :169.00
```

```
# 5
```

```
Days <- as.data.frame(seq.Date(from=as.Date("2010-01-01"),
                                to=as.Date("2019-12-31"),
                                by= "day"))

colnames(Days) <- "Date"
```

```
# 6
```

```
GaringerOzone <-left_join(Days,GaringerOzone)
```

```
## Joining with 'by = join_by(Date)'
```

```
dim(GaringerOzone)
```

```
## [1] 3652    3
```

Visualize

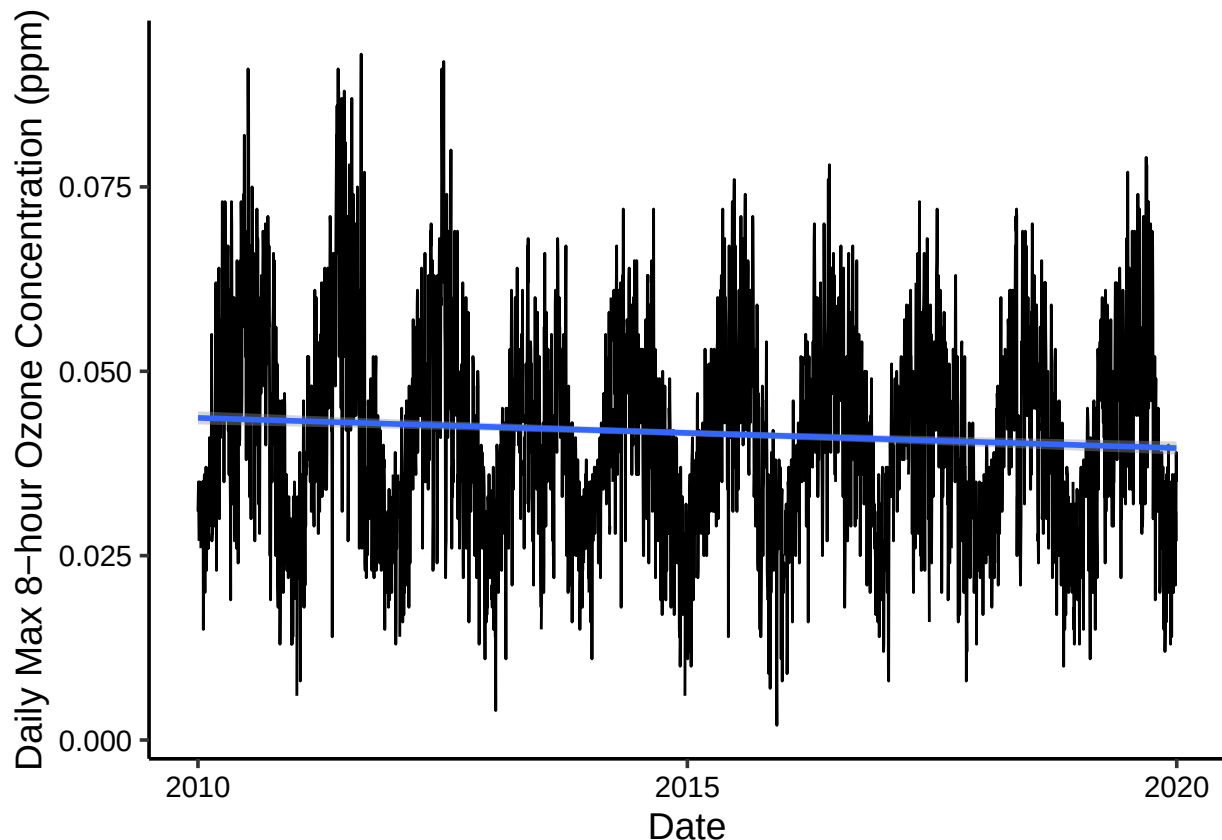
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
```

```
ggplot(GaringerOzone,
       aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth(method = "lm")+
  labs(
    x = "Date",
    y = "Daily Max 8-hour Ozone Concentration (ppm)")+
  mytheme
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: I see a slight downward trend in ozone concentration over time, suggesting a gradual decrease in levels from 2010–2019. I also see clear seasonal variability, with ozone concentrations generally higher in the summer months and lower in the winter.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

```
GaringerOzone_clean <-
  GaringerOzone %>%
  mutate(Daily.Max.8.hour.Ozone.Concentration.clean =
    zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))%>%
  select (Date, Daily.Max.8.hour.Ozone.Concentration.clean)

summary(GaringerOzone_clean$Daily.Max.8.hour.Ozone.Concentration.clean)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: We used linear interpolation because the ozone data show a generally negative linear trend over time and clear seasonal variation. Linear interpolation assumes gradual, steady changes between observations, which fits well for environmental data like ozone that vary across days and seasons. Using a piecewise constant (“nearest neighbor”) approach would hold each value constant until the next observation, creating a step-like pattern that ignores natural seasonal fluctuations and would make the series look unrealistic. As we learned in our lesson, spline interpolation would fit smooth curves between points, but this data does not require that level of smoothing it seems. Linear interpolation is a realistic and best fit way to fill in the missing values without compromising seasonal variance or overall trend. It also works well with datasets with small gaps like this one.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

#9

```
GaringerOzone_monthly <-
  GaringerOzone %>%
    mutate(Year = year(Date),
           Month = month(Date)) %>%
    group_by(Year, Month) %>%
    summarize(
      Month_Mean_Ozone = mean(Daily.Max.8.hour.Ozone.Concentration, na.rm = TRUE)
    )
```

‘summarise()’ has grouped output by ‘Year’. You can override using the
‘.groups’ argument.

```
GaringerOzone_monthly_final <-
  GaringerOzone_monthly %>%
    mutate(Date = my(paste0(Month, "-", Year))) %>%
    select(Date, Month_Mean_Ozone)
```

Adding missing grouping variables: ‘Year’

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

#10

```
f_month <- month(first(GaringerOzone_clean$Date))
f_year <- year(first(GaringerOzone_clean$Date))

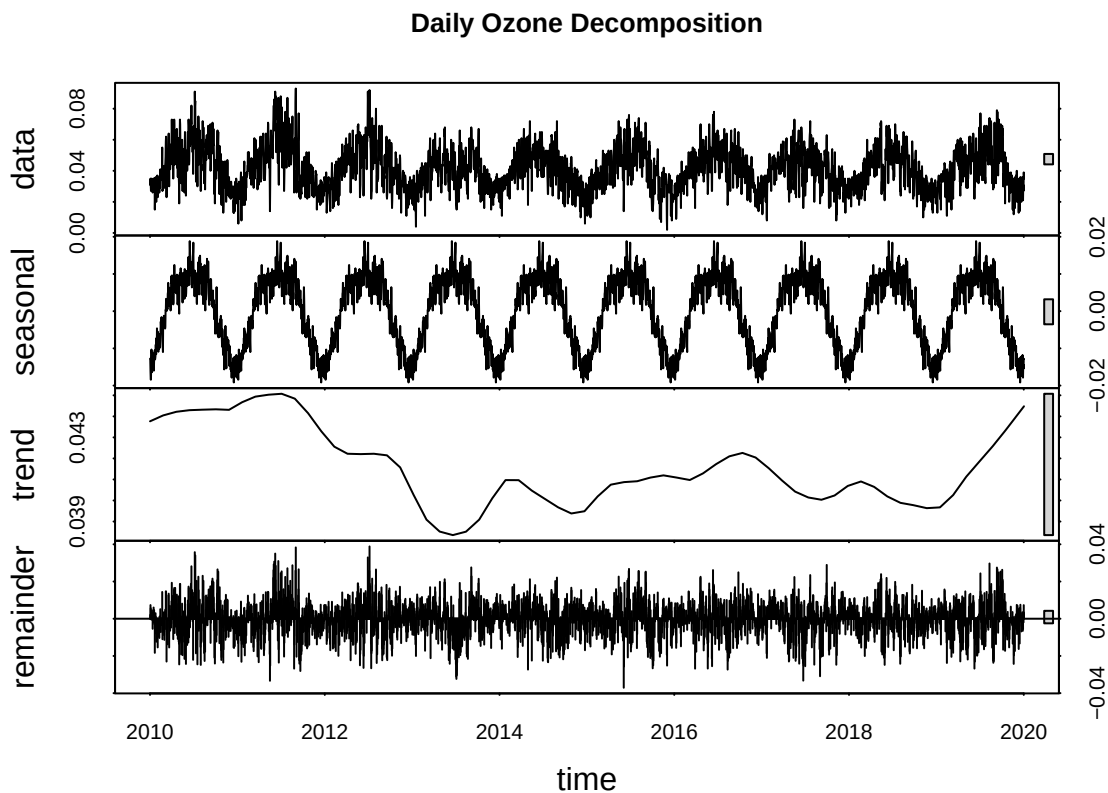
GaringerOzone.daily.ts <- ts(GaringerOzone_clean$
                             Daily.Max.8.hour.Ozone.Concentration.clean,
                             start = c(f_year, f_month),
                             frequency = 365)
```

```
f_month_m <- month(first(GaringerOzone_monthly_final$Date))
f_year_m <- year(first(GaringerOzone_monthly_final$Date))

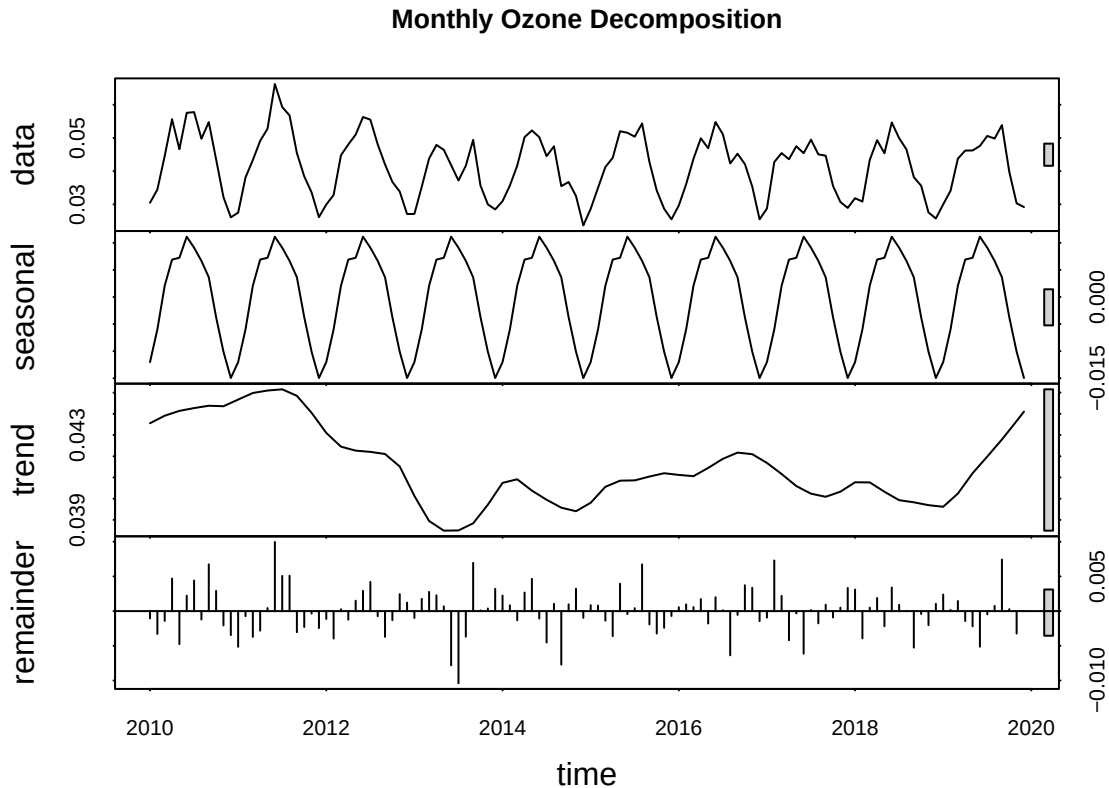
GaringerOzone.monthly.ts <- ts(GaringerOzone_monthly_final$Month_Mean_Ozone,
                              start = c(f_year_m, f_month_m),
                              frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
GaringerOzone_daily_decomp <- stl(GaringerOzone.daily.ts,s.window = "periodic")
plot(GaringerOzone_daily_decomp, main = "Daily Ozone Decomposition")
```



```
GaringerOzone_monthly_decomp <- stl(GaringerOzone.monthly.ts,s.window = "periodic")
plot(GaringerOzone_monthly_decomp, main = "Monthly Ozone Decomposition")
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12

```
GaringerOzone_monthly_data_trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
summary(GaringerOzone_monthly_data_trend)
```

```
## Score = -88 , Var(Score) = 1498
## denominator = 538.9944
## tau = -0.163, 2-sided pvalue =0.022986
```

Answer: The Seasonal Mann-Kendall test is the most appropriate test for this dataset because, as shown in the linear regression and decomposition, the data display clear seasonality. Since the ozone data is not normally distributed, a non-parametric test is more suitable. The Seasonal Mann-Kendall test is designed to detect a monotonic trend or, a consistent upward or downward trend, in data that is seasonal and unevenly distributed over time.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

13

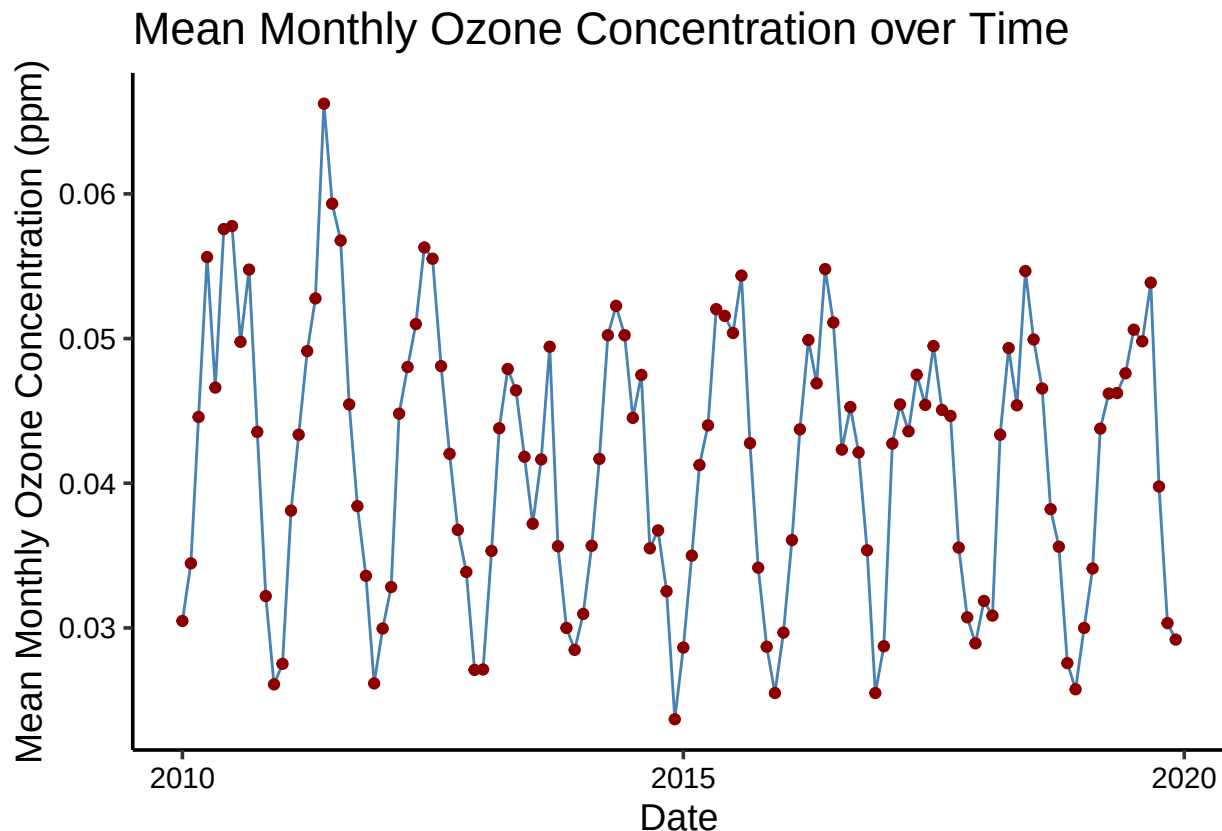
```
ggplot(GaringerOzone_monthly_final,
```



```

aes(x = Date, y = Month_Mean_Ozone)) +
  geom_line(color= "steelblue")+
  geom_point(color= "darkred") +
  labs(
    x = "Date",
    y = "Mean Monthly Ozone Concentration (ppm)",
    title = "Mean Monthly Ozone Concentration over Time"
  )+
  mytheme

```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: An analysis of mean monthly ozone concentrations confirms the research question: ozone levels at this station did change over the 2010s, showing a statistically significant downward trend. The graph demonstrates strong seasonal variability over time. The Seasonal Mann–Kendall test, which accounts for this seasonality, detected an underlying monotonic decrease in ozone levels (Kendall's tau = negative 0.163, p-value = 0.023).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
```

```
GaringerOzone_Seasonal_Component <-  
GaringerOzone_monthly_decomp$time.series[, "seasonal"]  
  
GaringerOzone_monthly_noseason_ts <-  
GaringerOzone_monthly.ts - GaringerOzone_Seasonal_Component
```

```
#16
```

```
GaringerOzone_monthly_data_trendnoseason <-  
Kendall::MannKendall(GaringerOzone_monthly_noseason_ts)  
summary(GaringerOzone_monthly_data_trendnoseason)
```

```
## Score = -1278 , Var(Score) = 194364.7  
## denominator = 7139  
## tau = -0.179, 2-sided pvalue =0.0037728
```

Answer: The standard Mann-Kendall test on the non-seasonal time series yielded a p-value of 0.0038. This is highly significant and even lower than the p-value from the seasonal test, suggesting the trend is very unlikely to be due to random chance. Furthermore, the tau value of negative 0.179 confirms the downward monotonic trend, reinforcing the conclusion that ozone levels have significantly decreased over the study period of the 2010s.